

LatchMoE: Graph-Compatible Expert Offloading through Address-Stable Residency

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Abstract

Sparse mixture-of-experts models bound per-token arithmetic but retain a large parameter footprint. Expert offloading addresses this footprint, yet a bounded expert cache changes weight placement as routing changes, conflicting with the stable storage bindings expected by graph replay. Our key insight is that replay requires stable graph-visible storage, not static logical expert residency. We present LatchMoE, which stages routed experts across an eager seam into fixed device slots, executes oversized working sets in capacity-bounded waves, and uses versioned leases to order transfer readiness, mapping publication, computation, and reuse. A vLLM-Ascend implementation evaluates the three target models under 11 semantic, graph, and lifecycle gates; Qwen3-Next’s native full-resident end-to-end sub-check is capacity-limited on this device. On Qwen3-30B-A3B, bounded waves change paired TTFT by -35.13% in an internal comparison against graph-compatible full-layer staging; its 95% CI is [-35.67%, -34.60%]. Enabling overlap in a one-factor campaign changes TTFT by -27.87%; its 95% CI is [-30.18%, -25.24%]. Performance results are scoped to one Ascend 910B2, BF16, tensor parallelism one, and serving concurrency one. Against full residency, the same workload trades a median sampled HBM peak from 97% to 80% for +261.28% TTFT and +309.51% TPOT; the reported TTFT reductions therefore characterize capacity-bounded staging rather than a universal end-to-end speedup.

1 Introduction

Mixture-of-Experts (MoE) architectures have emerged as a prominent approach for scaling Large Language Models (LLMs) to hundreds of billions of parameters [2, 5, 8, 16]. By dynamically routing each token to only a sparse subset of experts, MoE models substantially increase model capacity while keeping per-token computation bounded [3, 15]. However, sparse activation reduces computation rather than model storage: although only a few experts are executed for a given token, the weights of all experts must remain accessible because any expert may be selected at runtime. As MoE models scale, the complete expert pool can exceed accelerator High-Bandwidth Memory (HBM) capacity. To serve such models on memory-constrained accelerators, **expert offloading** has become an important strategy: the runtime keeps a portion of expert weights in host memory and accesses or stages them on demand during inference [7, 10, 25].

To make expert offloading practical, prior systems reduce its data-movement overhead through expert caching,

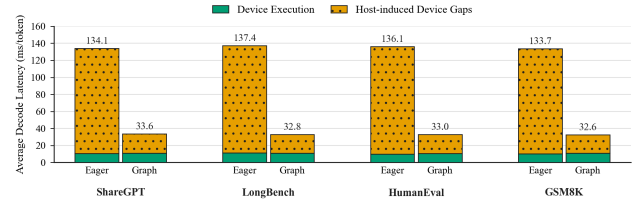


Figure 1. Average decode latency breakdown under eager execution and graph replay. Graph replay substantially reduces host-induced device gaps while leaving device execution time nearly unchanged.

routing-aware prediction and prefetching, and overlap between transfers and computation [1, 4, 7, 9, 25]. These techniques reduce the frequency of host-to-device transfers or hide part of their latency. However, they do not remove the execution dynamism of cache-based expert offloading: expert admission and eviction in a bounded HBM cache can change device storage assignments across inference steps.

Such runtime changes can be naturally accommodated under eager execution, where the host resolves the current device storage assignments and dispatches the corresponding kernels at each decoding step. This flexibility, however, comes at a substantial performance cost. Autoregressive decoding repeatedly executes a largely recurring sequence of fine-grained accelerator kernels, and launching these kernels individually incurs repeated host-side scheduling, dispatch, and synchronization overhead. **Graph replay** mitigates this overhead by capturing the recurring execution sequence once and replaying it as a unit, thereby reducing host-induced gaps between kernel executions [17, 26].

As shown in Figure 1, graph replay reduces average decode latency by approximately 75% across four workloads, from 133.7–137.4 ms/token under eager execution to 32.6–33.6 ms/token, while device execution time remains nearly unchanged. This result indicates that most of the improvement comes from eliminating host-induced device idle time rather than accelerating the underlying kernels.

However, graph replay requires the storage bindings referenced by captured computation to remain replay-stable. A dynamically managed expert cache can violate this requirement when routing-driven expert admission and eviction reassign logical experts to different device storage locations that are directly referenced by captured kernels. If such reassignment

changes the storage bindings observed by the captured computation, the runtime must either repair or recapture the graph to reflect the new bindings, or bypass replay and fall back to eager execution. These options reintroduce host-side intervention and diminish the benefit of graph replay. The resulting systems problem is to **support routing-driven expert replacement within a bounded HBM cache while keeping the storage bindings observed by replayed expert computation stable**.

Existing graph-compatible offloading mechanisms solve only parts of this problem. On CUDA platforms, UVA provides stable access to host-resident weights, but does not support selective expert replacement within a bounded HBM cache [13, 22]. Prefetch-based offloading can preserve graph replay by staging weights into preallocated device buffers [21]. Applied to MoE, however, layer-wise staging must move the entire expert layer despite sparse expert activation, incurring unnecessary HBM capacity and transfer costs. Exploiting this sparsity requires expert-granular staging, where the required experts become routing-dependent and may exceed the available device slots. Existing approaches therefore do not directly support routing-driven expert replacement over bounded, replay-stable HBM slots.

Our key insight is that **runtime dynamism need not change the execution state captured by the graph**. Graph replay requires stable storage bindings, not static expert residency. We therefore decouple logical expert identity from device storage: a fixed set of HBM slots remains bound to captured computation, while routing-dependent experts can dynamically occupy and leave those slots. This abstraction allows expert residency to evolve without changing the storage interface used by graph replay.

Realizing this abstraction raises three systems challenges. *Challenge 1: Routing-dependent expert placement must not change the storage bindings used by captured computation*. The experts required by a layer are revealed only after routing, yet routing-dependent residency decisions must not change the storage bindings or execution structure observed by captured expert computation. *Challenge 2: The routed expert working set can exceed the available HBM capacity*. Staging all required experts at once violates the memory budget, whereas fetching them sequentially exposes repeated weight-transfer latency on the inference critical path. *Challenge 3: Slot reuse must remain correct under asynchronous transfer and computation*. Reassigning a slot before its previous consumer finishes can overwrite weights still in use, while exposing a new assignment before transfer completion can make computation observe incomplete contents.

We present LatchMoE, a graph-compatible MoE offloading runtime that virtualizes routing-dependent expert ownership over a fixed set of replay-stable HBM slots. LatchMoE places expert staging and ownership transitions at explicit replay boundaries, allowing captured expert computation to

remain bound to the same slots while their logical owners change. When the routed working set exceeds slot capacity, LatchMoE partitions expert execution into multiple waves and pipelines the staging of subsequent waves with ongoing computation. Finally, versioned slot leases coordinate transfer readiness, computation completion, and ownership reassignment, preventing a slot from being reused before its previous lifetime ends or exposed before its new contents are ready.

In summary, this paper makes the following contributions:

- We identify that the conflict between dynamic expert offloading and graph replay arises when changing expert placement also changes the storage bindings referenced by captured computation. We show that graph replay requires stable storage bindings, not static expert residency, enabling dynamic experts to be virtualized over a fixed storage interface.
- We design LatchMoE, a graph-compatible MoE offloading runtime that virtualizes routing-dependent logical ownership over replay-stable HBM slots. LatchMoE manages expert staging and ownership transitions at explicit replay boundaries, supports working sets larger than slot capacity through multi-wave execution, and uses versioned slot leases to safely coordinate asynchronous transfer, computation, and slot reuse.
- We implement LatchMoE on vLLM-Ascend and validate its execution semantics across three MoE architectures, preserving exact outputs and graph replay while exercising dynamic residency, overflow, and asynchronous slot lifecycles. Performance claims remain scoped to a matched single-NPU Qwen3-30B-A3B configuration, where multi-wave execution reduces prefill TTFT relative to graph-compatible full-layer staging.

2 Background and Problem Formulation

2.1 MoE Inference and Cache-Based Expert Offloading

A Mixture-of-Experts (MoE) layer replaces the dense feed-forward network with a set of experts and a router. For each input token, the router selects a small subset of experts and combines their outputs according to the routing weights [2, 5, 8, 16]. We refer to an expert’s model-defined identity as its *logical expert ID*. For an invocation of a given MoE layer at step t , let $\mathcal{A}_t$ denote the set of logical experts selected across all participating tokens. Because routing is input-dependent, $\mathcal{A}_t$ varies across inference steps. This active-set variability is especially pronounced during prefill, where many prompt tokens are processed together and may collectively activate substantially more experts than a typical decoding step [4].

When the complete expert pool does not fit in accelerator HBM, expert offloading places part of the expert weights

in host memory. We focus on *cache-based expert offloading*, which maintains a bounded expert cache in HBM and stages non-resident experts from host memory on demand [1, 4, 7, 25]. Let $\mathcal{H}_t$ denote the experts of this layer currently resident in HBM, with $|\mathcal{H}_t| \leq C$, where C denotes the maximum number of experts that the configured cache can hold. A requested expert $e \in \mathcal{A}_t \cap \mathcal{H}_t$ is a cache hit, whereas $e \in \mathcal{A}_t \setminus \mathcal{H}_t$ incurs an expert cache miss and must be staged from host memory. If the cache is full, admitting a missing expert requires evicting or replacing an existing resident expert.

Cache-based offloading therefore makes expert residency and placement runtime-managed state. For each resident expert $e \in \mathcal{H}_t$, let $\pi_t(e)$ denote the device storage location assigned to its weight tensors at step t . As routing decisions trigger admission, eviction, and replacement, both the resident set $\mathcal{H}_t$ and the placement mapping π_t may change across inference steps. Thus, a logical expert has a fixed model identity, while its HBM residency and device storage location are determined dynamically at runtime.

2.2 Graph Replay and Stable Storage Bindings

Graph replay reduces repeated host-side scheduling and kernel-launch overhead by capturing a recurring accelerator execution sequence and replaying it as a unit [17, 26].

Captured computation reuses both its operation structure and the storage bindings referenced by those operations. We denote these storage bindings by $\mathcal{B}$.

Graph replay does not require the contents of this storage to remain unchanged. Values in preallocated buffers may be updated between replays as long as the captured computation continues to access the same storage locations. For example, FlashInfer updates request-dependent scheduling metadata in preallocated device workspaces while preserving the workspace locations referenced by captured kernels [26].

Thus, graph replay can accommodate dynamic data behind stable storage bindings. It is a change to the bindings themselves that may require graph update or recapture [14].

2.3 Dynamic Expert Placement Meets Graph Replay

The conflict arises when cache-based expert offloading exposes the current device storage assignment of expert weights directly to captured computation. Consider a logical expert e that is resident at two inference steps t and $t' > t$. If e is evicted after step t and later reloaded, it may occupy a different HBM location:

$$\pi_t(e) \neq \pi_{t'}(e).$$

Such placement changes are natural under a bounded expert cache, whose contents evolve with routing decisions.

Dynamic expert placement does not inherently conflict with graph replay. The conflict occurs when captured computation directly references the current storage location of a logical expert.

If captured expert computation directly binds e to its current HBM location, a change in $\pi_t(e)$ also changes the graph-visible binding $\mathcal{B}$. Dynamic expert caching may therefore require the resident set $\mathcal{H}_t$ and placement mapping π_t to evolve, while conventional graph replay expects its graph-visible bindings $\mathcal{B}$ to remain reusable. Cache-based expert offloading and graph replay therefore impose different requirements on storage state:

$$(\mathcal{H}_t, \pi_t) \text{ may evolve, } \mathcal{B} \text{ remains replay-stable.}$$

The desired abstraction must allow changes in expert residency and placement without propagating those changes to the storage bindings observed by captured computation.

Several straightforward alternatives avoid one side of this conflict only by sacrificing another system objective. Keeping all experts permanently resident would stabilize their physical bindings but defeats offloading under a bounded HBM budget. Falling back to eager execution accommodates arbitrary placement changes but forfeits the host-overhead savings of graph replay. Updating or recapturing the graph after each placement change similarly reintroduces host intervention into the repeated inference path.

Therefore, our goal is to support dynamic expert replacement within a bounded HBM cache while preserving the graph-visible storage bindings required by graph replay.

3 Empirical Motivation

The conflict between dynamic expert offloading and graph replay is fundamental, but its practical severity remains an empirical question. If routing rarely changes residency, or if typical working sets fit in bounded HBM, or if transfer latency is negligible, simpler solutions would suffice. We measure three production MoE architectures spanning 30B to 80B parameters under realistic serving conditions to establish whether the problem demands a new system design.

All measurements use one Ascend 910B2, BF16, tensor parallelism one, disabled prefix caching, and serialized requests (both concurrency controls set to one).¹ We profile 200, 100, and 64 ShareGPT requests on Qwen3, GLM, and Qwen3-Next, respectively, generating at most 128 output tokens for Qwen3 and GLM, and 32 for Qwen3-Next to bound collection cost. An independent audit verifies event-stream integrity, balanced decode sampling, and deterministic reproducibility of all reported statistics. Figure 2 summarizes the key observations.

3.1 Expert Residency Changes Frequently Across Requests

A 32-expert cache serving Qwen3, GLM, and Qwen3-Next experiences cache misses on 20.6%, 14.0%, and 38.2% of decode expert accesses despite sparse activation (8, 4, and 10

¹Qwen3-30B-A3B: 48 routed layers, 128 experts/layer, top-8; GLM-4.7-Flash: 46 routed layers, 64 experts/layer, top-4; Qwen3-Next-80B-A3B-Instruct: 48 routed layers, 512 experts/layer, top-10.

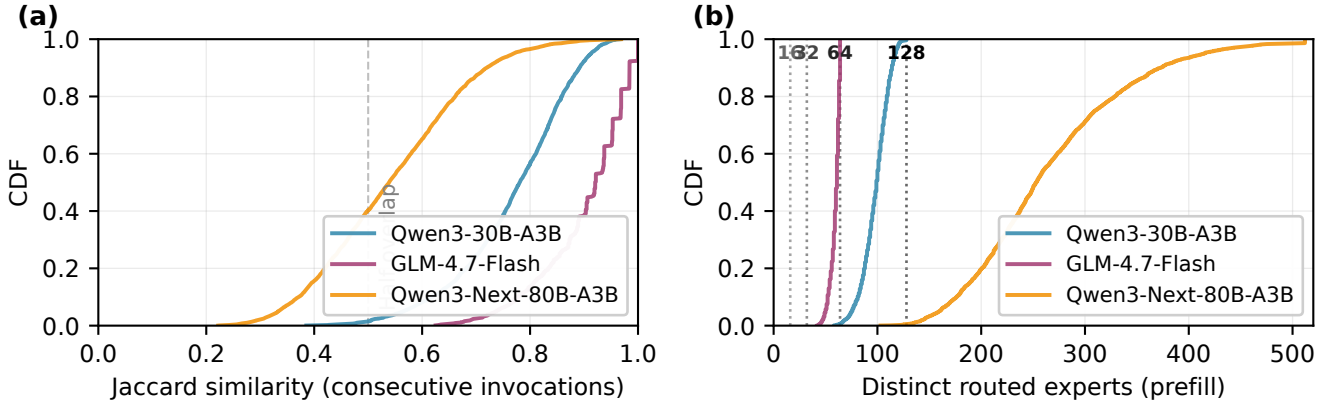


Figure 2. Empirical motivation for graph-compatible expert offloading. (a) CDF of Jaccard similarity between consecutive prefill invocations within each MoE layer, showing substantial expert turnover across requests. Even moderate similarity (e.g., 0.5–0.8) requires frequent mapping updates, as 20–50% of experts change between invocations. (b) CDF of distinct routed experts per prefill invocation. Vertical lines mark cache capacities; all three models exhibit working sets that substantially exceed practical capacity bounds.

experts per layer). More critically, 68.7%, 41.0%, and 94.2% of layer invocations trigger at least one logical-to-physical mapping update.

Figure 2a quantifies expert turnover across consecutive prefill invocations within each layer. While Qwen3 and GLM exhibit median Jaccard similarities of 0.78 and 0.92, Qwen3-Next shows substantially lower overlap at 0.54. Even moderate similarity values translate to frequent residency changes: a Jaccard similarity of 0.78 means 22% of experts differ between consecutive invocations, and 0.54 means 46% turnover. For Qwen3-Next, over 60% of consecutive invocations share fewer than two-thirds of their experts.

This turnover directly challenges systems that bind captured computation to expert-specific storage locations. Although sparse activation limits per-token computation, it does not stabilize expert residency. Each mapping change would require graph recapture under naive storage bindings, negating the host-overhead savings graph replay provides (Figure 1 in Section 1).

Implication. Graph-compatible offloading must decouple logical expert identity from graph-visible storage bindings. Replay requires stable addresses, not static residency.

3.2 Working Sets Exceed Practical Cache Capacity

Prefill aggregates routing decisions across an entire prompt. Figure 2b shows the distribution of distinct routed experts per prefill invocation, with vertical lines marking cache capacities of 16, 32, 64, and 128 slots. For a 32-slot cache, *all* 2,400 Qwen3 prefills, *all* 1,100 GLM prefills, and *all* 3,072 Qwen3-Next prefills overflow. The median active expert counts are 99, 61, and 253; the 95th percentiles reach 117, 64, and 415; maxima span 128, 64, and 512.

Even aggressive capacity scaling provides limited relief. A 64-slot cache still overflows 99.3% of Qwen3 prefills and every Qwen3-Next prefill; a 128-slot cache still overflows 99.5% of Qwen3-Next prefills. The corresponding median active-weight footprints are 0.87, 1.07, and 1.48 GiB, versus 32-slot budgets of 0.28, 0.56, and 0.19 GiB. Overflow occurs with a single in-flight request—concurrent batching is not a prerequisite.

Implication. A bounded HBM cache must support intra-invocation slot reuse. Wave-based execution partitions the routed working set while preserving every token–expert pair and its original routing weight.

3.3 Transfer Latency Justifies Overlap

Multi-wave execution exposes substantial data movement. Across profiled prefills, Qwen3, GLM, and Qwen3-Next execute 8,571, 2,200, and 26,695 waves, transferring 1.58, 0.60, and 4.62 TB from host memory. Even with prefetching, aggregate staging wait accounts for 20.2%, 24.0%, and 17.2% of the corresponding MLP computation time.

Implication. Transfer–compute overlap is essential for practical performance. Sequential wave execution would expose staging latency on every prefill’s critical path. However, asynchronous overlap requires careful coordination between transfer readiness, computation completion, and slot reuse. Section ?? describes our versioned lease protocol to ensure correctness when these events advance on independent streams.

These observations establish three requirements: stable storage bindings despite frequent remapping (§??), bounded-capacity execution despite oversized working sets (§??), and safe asynchronous overlap despite concurrent slot reuse (§??).

4 Design

4.1 Overview

LatchMoE’s central principle is that *logical expert ownership may change, while graph-visible storage remains fixed*. For each managed layer, it preallocates bounded expert slots and a persistent logical-to-slot map. Their allocations survive capture and replay; routing changes only slot contents and map values.

Figure 3 shows the resulting captured-router, eager-stage, and captured-MLP path. Staging transfers misses and publishes the map before the MLP resumes through the same graph-visible interface.

Three mechanisms realize this model: replay-stable slots decouple residency from addresses; pipelined waves reuse bounded storage; and versioned leases order transfer, publication, computation, and reuse.

4.2 Capability-Driven Semantic Contract

Capability descriptor. Before selecting the seam, LatchMoE materializes a serializable descriptor for each managed layer. The descriptor records (i) the output contract; (ii) the shared-expert representation and activation mode; (iii) whether routing logits are externally supplied or produced by an internal gate; (iv) all top- k , grouping, scoring, bias, normalization, and scaling parameters; (v) the owner of final combination; and (vi) weight, parallelism, and overlap modes. The selection rule uses this tuple rather than a model allowlist. Routed-only and externally shared-expert layers select distinct, fixed return schemas, avoiding a data-dependent captured type.

Fail-closed selection. The seam is enabled only when every descriptor field belongs to an implemented profile. Unknown router adapters, unregistered or mismatched fused shared/routed placement, quantized weights, multi-NPU execution, and shared-expert overlap are rejected before capture with the full descriptor and blocker list; rejection never silently chooses an eager or monolithic path. Manifests therefore identify the exact accepted tuple.

4.3 Replay-Stable Dual-Lane Storage

Routed experts form a dynamic lane. For each managed layer, LatchMoE retains their W_{13} and W_2 weights in a CPU-first host store and allocates two contiguous device tensors containing a bounded number of physical expert slots. A persistent tensor maps each logical routed expert ID to a slot ID. The slot tensors and mapping allocation survive capture and replay; replacement changes only slot contents and mapping values.

The shared lane remains device-resident because it contributes to every applicable token; caching it would consume capacity without sparsity and repeat shared computation in each overflow wave. Its bytes are charged separately in the

memory ledger, while capacity, victim, and wave decisions cover only routed experts.

Each routed slot records a logical owner, a monotonically increasing version, a state, and its last-use step. A *lease* is the tuple (slot ID, logical expert, version). The version distinguishes ownership intervals that reuse the same physical address. The central storage invariant is therefore stronger than pointer stability: captured computation may consume a routed slot only through a mapping published for a ready lease, while the shared lane retains a fixed owner and allocation throughout replay.

4.4 Semantics-Preserving Router Decomposition

The staging decision is known only after routing, yet a monolithic fused-MoE operation hides expert selection from the graph partitioner. LatchMoE therefore relocates, rather than reimplements, the selection step into a captured router operation. For an external-logit layer, the operation consumes the original router logits. For an internal-gate layer, it first invokes the same gate on the same hidden states. It then calls the native selector with the layer’s top- k , grouping, scoring function, renormalization flag, correction bias, routed scaling factor, and logical expert count.

The router emits the original top- k IDs and weights. The eager boundary uses only IDs to plan residency and cannot edit either tensor. Captured expert computation injects both into the native apply path, so selection occurs once. The seam relocates routing without approximating its mathematical decision.

Output semantics remain native: the routed-only contract returns one tensor, whereas the shared-expert contract returns the expected fixed pair. The native runner retains shared gating, routed scaling, output transforms, and final addition; LatchMoE owns only residency and ordering.

4.5 Replay-Boundary Staging and Safe Publication

After the captured router completes, the eager staging boundary deduplicates the routed IDs and asks a placement policy for an ordering. The returned plan must be a permutation of the actual routed set: it may prioritize a hit, predicted reuse, or transfer cost, but it cannot add an unrouted expert or omit a routed expert. Hits retain their current leases. For a miss, the transfer engine selects a reassignable slot, increments its version, installs the new logical owner, copies W_{13} and W_2 on a dedicated H2D stream, and records a readiness event.

Publication follows a strict happens-before order. The runtime first checks that the slot still matches the issued lease. It then makes the consumer stream wait for the readiness event, transitions the slot to ready, and only afterward updates the persistent logical-to-slot mapping. Publishing earlier would allow captured compute to observe either partially transferred weights or a completion event from an old owner. Hits and misses leave the boundary through the same

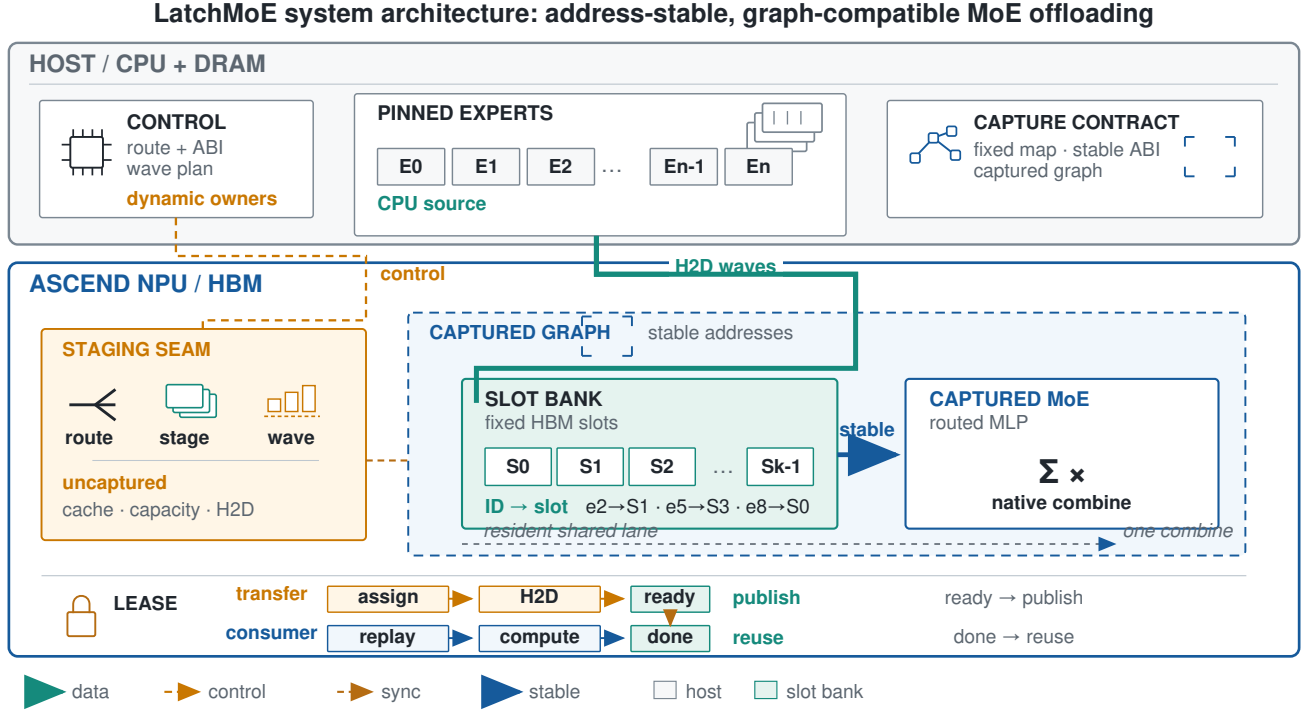


Figure 3. LatchMoE separates dynamic control from graph-visible data. The host control plane computes routing and wave plans while pinned expert weights remain outside the captured region. At the replay boundary, staging checks capacity and transfers misses; the captured NPU region sees only the fixed slot bank, its logical-to-slot map, and stable addresses feeding the fused routed-MoE kernel. The versioned lease at the bottom permits overlap while publishing only after a transfer is ready and reusing a bank only after its consumer lease is done.

address-stable interface, so the subsequent graph piece does not encode how a routed expert became resident.

The stage is an explicit graph split point: capture fixes the router and MLP pieces while permitting routing-dependent host work between them. Pointer checks cover the slot banks and map; the descriptor fixes the operator schema.

4.6 Versioned Transfer-Compute Lifecycle

Address stability alone does not make asynchronous reuse safe. A slot moves through four externally visible states: *empty*, *loading*, *ready*, and *computing*. Only empty or ready slots may be selected for a new owner. Before routed computation begins, the runtime acquires leases for all referenced slots and marks them computing. A compute event protects those leases until the consumer stream finishes; release again verifies owner and version before returning each slot to ready.

Two temporal invariants follow: version validation prevents a stale transfer from publishing, and excluding computing slots prevents in-flight overwrite. The same lease protocol protects transient prefill stage banks: bank b cannot receive wave $k + 2$ until the compute event for wave k has

released its leases. Lifecycle violations are surfaced as run failures rather than being hidden by fallback.

More precisely, for lease $\ell = (s, e, v)$, replay requires the address of slot s to remain constant. Publishing ℓ requires $\text{ready}(s, e, v)$; reassigning s at version $v + 1$ requires $\text{done}(s, e, v)$. Owner/version validation binds both events to the same ownership interval, so unrelated completions cannot authorize publication or reuse.

4.7 Shared-Aware Capacity-Bounded Composition

Prefill can route a batch of tokens to more distinct routed experts than fit in the slot budget. Let P be the flattened set of routed token-expert pairs, including each pair’s original top- k position and weight. LatchMoE partitions P into waves $P_1, \dots, P_m$ such that each wave requires no more routed experts than a stage bank can hold. Pair membership and routing weights are immutable. Hit-only work may execute from persistent slots, while miss waves use transient banks governed by the same lease rules.

Each wave computes routed outputs while preserving pair positions. After all waves, LatchMoE constructs one global native unpermute index and invokes routed combine once with the original weights and shape. Coverage validation

rejects missing or duplicate pairs; single recombination preserves reduction and model-specific postprocessing semantics.

Shared work remains outside the partition and is combined once with the fully reconstructed routed contribution. With multiple transient banks, transfer may populate wave $k + 1$ while wave k computes without changing routing or final combination. The storage cost is bounded; Section 6 measures whether compute hides transfer.

4.8 Policy Boundary and Qualification Scope

The data plane exposes only a constrained placement-plan interface: layer ID, complete routed set, and a valid ordering. The default policy preserves first-seen order; another policy may reorder staging but must obey the same capacity, publication, and lifecycle invariants. This boundary lets policy studies reuse one execution substrate without making prediction quality part of the correctness argument.

The design covers single-NPU unquantized BF16 execution, with either no shared expert, an externally represented resident shared lane, or a registered fused shared suffix, and supported external-logit or internal-gate selectors. This ABI alone does not establish correctness or performance for every accepted model; each tuple still requires router/boundary parity, exact outputs, replay evidence, and matched runs. Section 6 reports the qualified boundary.

5 Implementation

We implement LatchMoE as a Python package that registers through vLLM’s general plugin entry point. The package integrates with a hook-enabled vLLM-Ascend runtime; the upstream request scheduler and OpenAI-compatible serving API remain unchanged. The host runtime and the Ascend platform plugin are pinned as a compatible stack because the staging boundary changes the internal fused-MoE execution contract rather than the public serving interface.

5.1 Graph integration

The integration decomposes a managed MoE layer into a captured router, the custom `vllm: :moe_offload_stage` splitting operation, and a captured expert MLP. The splitting operation returns control to the host after routing so that LatchMoE can stage the active experts. The subsequent graph piece reads the persistent slot weights and logical-to-slot buffer. The runtime locks their data pointers at capture and validates them before replay.

Graph-compatible runs use PIECEWISE ACLGraph. The benchmark validator requires explicit capture and replay records and rejects forced eager execution, eager fallback, graph errors, address changes, stale H2D completion, and slot-version violations. Disabling LatchMoE restores the host runtime’s null hooks. The launcher also rejects simultaneous use of the native weight-offload path and LatchMoE because

two independent owners of expert storage would violate the slot lifecycle.

5.2 Loading and memory management

CPU-first loading creates managed expert tensors in host memory during model initialization. The host store keeps one contiguous W_{13} tensor and one contiguous W_2 tensor per layer and exposes per-expert views to the transfer engine. It checks shape, dtype, stride, device, layer coverage, and expert coverage before releasing the original device expert banks. Host pinning is used when the runtime supports it and reports failures explicitly.

Automatic configuration derives the resident-layer selection and slot count from the requested offload amount, physical HBM, a KV-cache reserve, and a cap on the fraction of HBM assigned to slot storage. Explicit overrides remain available for controlled sensitivity experiments. The memory ledger accounts separately for the host store, persistent slots, transient prefill stage banks, mapping tensors, full-layer fallback storage, and any original weights that remain allocated.

5.3 Transfers and lifecycle enforcement

The transfer engine maintains a dedicated H2D stream. It validates layout compatibility before copying and batches consecutive host and slot slices when their storage is contiguous. Each asynchronous load returns a readiness object that contains the NPU event and the exact slot leases made consumable by that event. The consumer installs the event dependency before the object marks the leases ready. The runtime then publishes mapping values.

Compute protection uses an event recorded after the expert operation. Slots remain in the computing state until the event completes and every recorded lease still matches its current owner and version. Shutdown drains pending transfer and compute work, closes profiling writers, releases managed storage, and emits a release acknowledgement for the benchmark harness.

5.4 Wave execution and profiling

The prefill path supports a configurable number of stage banks and prefetch depth. The qualified configuration uses two stage banks and depth one. Wave plans carry the original pair offsets needed for one native recombination, and strict validation rejects duplicate or missing coverage. A preflight-qualified overflow decision may select the explicit full-layer mode; NPU, ACL, memory, address, semantic, and lifecycle failures are not swallowed by that path.

Profiling records JSONL events for layer registration, slot ownership, transfers, mapping publication, wave staging and compute, graph replay issue, memory accounting, and service release. Benchmark manifests bind each run to the runtime revisions, model and workload digests, device, command line, environment, graph mode, and raw client/server

artifacts. These records support the fail-closed checks used in Section 6.

6 Evaluation

We ask six questions. **Q1** tests semantic fidelity, graph replay, and the lifecycle contract across the three target models. **Q2** establishes baseline feasibility and quantifies the cost of bounded offloading relative to full residency. **Q3** measures capacity-bounded waves against graph-compatible full-layer staging. **Q4** isolates transfer–compute overlap. **Q5** varies slot capacity. **Q6** accounts for storage, transfer, and unsupported boundaries.

6.1 Experimental Methodology

Platform and software. Experiments use one Ascend 910B2 with 65,536 MiB HBM and tensor parallelism one, hosted by a 192-core Kunpeng-920 server with 2.0 TiB DRAM. Models use BF16 weights. The managed stack pins PyTorch 2.10.0, torch-npu 2.10.0.post2, vLLM commit ad7125a, and vLLM-Ascend commit 4806367. Each run manifest records the complete command, model and dataset digests, source-tree digest, environment, device, graph mode, and service-release custody.

Workloads and controls. The matched baseline, overlap, and capacity campaigns draw from one frozen 64-request ShareGPT prefill-heavy bucket with 1034–1979prompt tokens (median 1394), generate 32 tokens at temperature zero and seed 42, disable prefix caching, and set both client concurrency and `max_num_seqs` to one. The full-layer/multi-wave campaign uses a separate frozen 200-request ShareGPT manifest and 128 output tokens. Every formal unit starts a fresh service. Counterordered comparisons use three start pairs; the paired estimator is the exponential of the mean, across starts, of the median per-request log latency ratio. We report a 10,000-replicate hierarchical bootstrap with seed 20260823: 95% intervals for TTFT effects and 90% intervals for the preregistered TPOT equivalence test (margin $\pm 5\%$). For the Qwen3 performance campaigns, the 14-GiB Auto-Config manages 12 of the model’s 48 MoE layers (one layer in each four-layer group); all four baseline arms and Q3–Q5 use this identical layer selection. Q1 is a separate capability qualification and exercises the model-specific layer counts shown in Table 1.

Fail-closed gates. A performance unit must complete all requests, match complete output-token arrays by request ID, capture and replay PIECEWISE ACLGraph, retain fixed slot and map addresses, publish no stale transfer, violate no owner/version lease, avoid eager fallback and NPU/ACL errors, emit a release acknowledgment, and retain raw client, server, profile, and HBM artifacts. A completed arm that fails exact-token comparison is retained but excluded from latency comparison.

6.2 Q1: Does LatchMoE preserve semantics and replay?

Table 1 covers routed-only Qwen3-30B-A3B, shared-expert GLM-4.7-Flash, and shared-expert Qwen3-Next-80B-A3B-Instruct. Each model is evaluated against the same 11 gate categories: native router parity, native layer-boundary parity, a native oracle where feasible, eager diagnostic parity, exact end-to-end tokens, graph capture and replay, zero eager fallback, overflow composition, decode cache churn, and H2D lease/release ordering. Qwen3-Next’s native full-resident end-to-end subcheck is capacity-limited on this device; its native semantics are therefore checked at the router and layer boundary, while its end-to-end comparison uses the eager seam as the reference. The router and layer-boundary checks compare four records per managed layer. Across all three capability tuples, every captured slot/map address validates at replay, every managed layer releases its original expert bank, and the online profiles exercise both overflow waves and H2D leases. Thus, the seam preserves the evaluated native routing and combination contracts rather than merely producing a runnable graph.

This is a capability qualification, not workload breadth: each model uses one short request in each end-to-end mode (four generated tokens) plus a one-token overflow stress request. Qwen3 and GLM compare native, LatchMoE-eager, and LatchMoE-graph outputs. Qwen3-Next cannot materialize its native full-resident path on this 64-GiB device, so its end-to-end gate compares eager-seam and graph outputs; native semantics remain checked at the router and layer boundary.

To expose the graph effect separately from the later multi-wave and overlap comparisons, we also ran a matched one-request qualification diagnostic on Qwen3 with the same 32-slot, 14-GiB configuration. The eager seam and PIECEWISE graph modes produced identical four-token output IDs; the observed TTFT was 2235.66 ms versus 2165.77 ms, and TPOT was 262.04 versus 140.63 ms/token. The diagnostic recorded 48 slot-address validations in each mode and 97 non-synchronizing PIECEWISE replay events. This is direct graph-versus-eager evidence for the qualified path, but it is one request and therefore is not a repeated-start variance estimate, throughput result, or concurrency claim.

We also established Slot+Eager as the primary causal baseline for this graph question and Slot+PIECEWISE as its only matched treatment. Across the frozen standard workload, we completed 24 fresh units: three counterordered starts for each of LongBench (200 requests), ShareGPT (200), HumanEval (164), and GSM8K (200), with identical model, slot, offload, KV-cache, prefix-cache, request-order, and sampling controls. This was a four-token qualification rather than a performance campaign. The audit records paired exact-token exceptions in ShareGPT, HumanEval, and GSM8K, while all LongBench requests exceed the current context

Table 1. Three-model semantic and graph qualification. $L/E/S$ denotes managed MoE layers, routed experts per layer, and resident shared experts. Router/boundary counts are exact native comparisons; locks/replays are address locks and observed graph replays; overflow/H2D counts exercise wave composition and transfer leases. Every row passes all 11 gates, including exact end-to-end tokens and zero eager fallback.

Model	$L/E/S$	top- k	Router	Boundary	Locks/replays	Overflow/H2D	Result
Qwen3-30B-A3B	48/128/0	8	192	192	48/97	144/239	Pass
GLM-4.7-Flash	46/64/1	4	184	184	46/94	92/226	Pass
Qwen3-Next-80B-A3B	48/512/1	10	192	192	48/97	144/240	Pass

boundary. Thus the baseline choice is fixed, but this campaign supports neither a four-dataset exact-equivalence claim nor a Graph latency superiority claim; its failed gates remain visible in the evidence ledger.

6.3 Q2: What baselines are feasible, and what does offloading cost?

We run three counterordered starts of four ACLGraph arms with an explicit 512-MiB KV cache and 32requests per unit: full-resident vLLM-Ascend, its native weight-prefetch path, the legacy layered offload path, and LatchMoE with a 14-GiB target. Full residency and LatchMoE each pass 96/96 exact requests. Native prefetch and legacy layered offload both complete 96/96 requests and capture/replay the graph, but respectively differ from the oracle on 30 and 30 request-token arrays across their three starts. We therefore retain both as unsupported comparison cells and draw no performance conclusion from them.

Full residency is a cost anchor, not a memory-matched offload baseline. Its repeat-median TTFT p50 and TPOT p50 are 179.13 ms and 13.19 ms/token, versus 662.22 ms and 54.50 ms/token for LatchMoE. The paired TTFT effect is +261.28% (95% CI [+240.09%, +277.32%]); the TPOT effect is +309.51% (95% CI [+299.13%, +319.38%]). This is the cost of host-backed expert replacement in the evaluated implementation. In return, the median sampled HBM peak falls from 97% to 80%, while exact output and graph replay remain qualified.

Table 2 makes the deployment trade-off explicit under the same matched Qwen3 contract. LatchMoE stores 13.50 GiB in the host expert store and 3.38 GiB in persistent slots, plus a 0.56 GiB prefill stage bank; the profile records 292.05 GiB of H2D traffic. The median sampled peak HBM falls by 17 percentage points, but TTFT p50 and TPOT p50 increase by 261.28% and 309.51%, respectively. Thus the result is a latency cost paid for single-card deployability, not a universal performance improvement.

6.4 Q3: Do bounded waves improve full-layer staging?

The graph-compatible full-layer arm materializes the layer working set before compute; the multi-wave arm is identical except that bounded banks execute and recombine the

Table 2. Same-scope full-resident cost anchor for Qwen3-30B-A3B: three fresh starts, 32 requests per arm, 512-MiB KV cache, and disabled prefix caching. HBM reclamation is the difference in median sampled peak utilization; stage wait is reported from the profile, not substituted for end-to-end latency.

Metric	Full resident	LatchMoE
TTFT p50 (ms)	179.13	662.22
TPOT p50 (ms/token)	13.19	54.50
Output throughput (tok/s)	52.94	13.23
Sampled peak HBM (%)	97	80
Host store (GiB)	0	13.50
Slot HBM (GiB)	0	3.38
Prefill stage HBM (GiB)	0	0.56
H2D traffic (GiB)	0	292.05
Prefill stage wait (ms)	0	257.11
Decode ready wait (ms)	0	95.05

Derived cost: LatchMoE reclaims 17 percentage points of sampled peak HBM, while adding +261.28% TTFT-p50 and +309.51% TPOT-p50 relative to full residency.

routed work in waves. Three AB/BA/AB pairs each serve 200 requests. All six units complete 1200 requests, match all 153600 output token IDs, capture and replay, and record zero fallback or lifecycle error.

TTFT changes by -35.13% with a 95% CI of [-35.67%, -34.60%]. As a conventional summary, the median across starts of TTFT p50 falls by 35.21%, and TTFT p95 falls by 22.96%. TPOT’s paired effect is 0.25% with a 90% CI of [-1.14%, 1.68%], entirely within the preregistered $\pm 5\%$ equivalence margin. The supported result is therefore a prefill-path TTFT reduction without a material TPOT change under this contract.

6.5 Q4: Does overlap contribute causally?

We next hold the wave plan, two stage banks, graph mode, slot capacity, workload, seeds, and source digest fixed, changing only prefetch depth from zero (serial staging) to one (overlap). Three AB/BA/AB pairs serve 64 requests per unit; all 384 requests pass exact-token, graph, lifecycle, and release gates. Overlap changes TTFT by -27.87% (95% CI [-30.18%, -25.24%]); the interval is entirely below zero and passes the

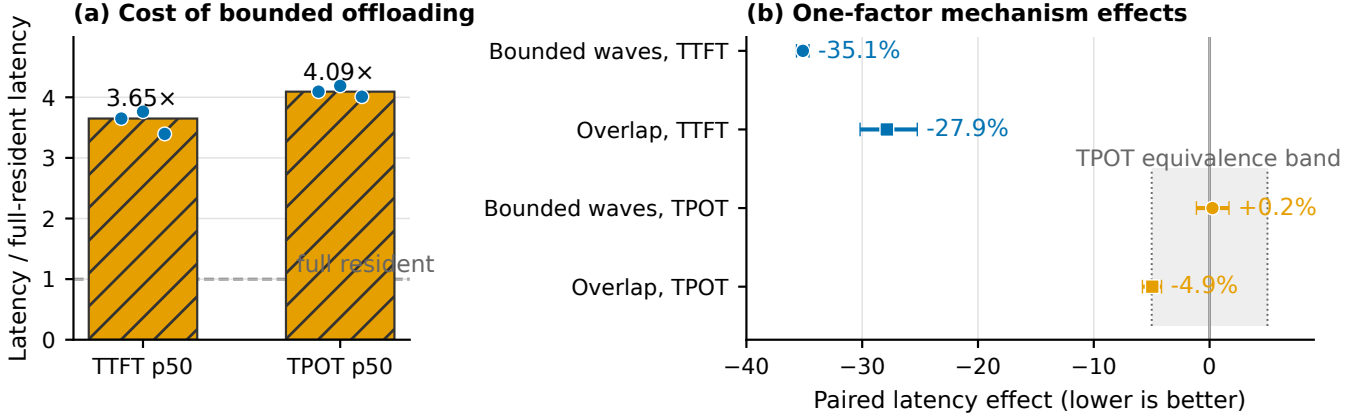


Figure 4. Bounded offloading has an explicit full-resident cost, while its two prefill mechanisms reduce exposed latency. (a) Bars are medians and points are three fresh exact-token, counterordered starts (96 requests per arm); the full-resident reference is 1.0. Native-prefetch and legacy-offload arms are omitted because they fail the exact-token comparability gate. (b) Bounded waves compare multi-wave against full-layer staging (200 requests per unit); overlap compares prefetch depth one against serial depth zero (64 requests per unit). Each contrast uses three counterordered pairs. Points are paired log-ratio estimators; whiskers are 95% CIs for TTFT and 90% CIs for TPOT. Lower is better. All experiments use Qwen3-30B-A3B, one Ascend 910B2, and concurrency one.

preregistered causal-benefit gate. TPOT changes by -4.95% (90% CI [-5.80%, -4.15%]). Its equivalence gate does *not* pass because the lower bound extends beyond -5%; we report the measured decrease rather than calling TPOT unchanged. This one-factor experiment supports overlap’s causal contribution, while Q3 measures the complete multi-wave path.

6.6 Q5: How does slot capacity change behavior?

We sweep 16, 32, and 64 physical slots with the same model, 14-GiB target, 512-MiB KV cache, first 32 manifest requests, and runtime source digest. Each point is one fresh start and passes exact-token, graph, capacity, lifecycle, and release gates; latency is therefore descriptive rather than an uncertainty claim. Figure 5 reports the resulting TTFT, average wave count, and persistent-slot storage. From 16 to 64 slots, TTFT p50 falls from 679.36 to 452.67 ms (33.37%), and p95 falls from 754.92 to 529.06 ms (29.92%). Total waves fall from 2808 to 792 and H2D volume from 318.5 to 173.8 GiB, while the persistent slot bank grows from 1.69 to 6.75 GiB and sampled peak HBM rises from 77% to 87%. This point quantifies the intended reuse-storage tradeoff; it is not a cross-workload scaling claim.

6.7 Q6: What resources and boundaries are explicit?

Table 3 accounts for each model’s graph-qualified 32-slot run. Host storage scales with the managed expert pool, while device storage depends on expert shape, shared residency, and stage banks rather than total expert count alone. Slot/stage storage is 13.5/0.56 GiB for Qwen3, 25.9/0 GiB for GLM, and 9.0/0.38 GiB for Qwen3-Next; mapping buffers are at most 0.19 MiB. Host-control time is 10.06 s, 0.073 s, and 44.85 s,

Table 3. Audited resources in each model’s graph-qualified LatchMoE run. Device storage sums the 32-slot bank, prefill stage banks, and resident shared-expert weights; H2D is the profiled transfer volume for that qualification workload. Original managed expert banks are released in every row.

Model	Host (GiB)	Device (GiB)	H2D (GiB)	Replays
Qwen3-30B-A3B	54.0	14.1	65.5	97
GLM-4.7-Flash	51.8	26.7	22.3	94
Qwen3-Next-80B-A3B	144.0	9.7	152.6	97

respectively. These workload-specific values are costs, not a cross-model ranking. Qualification profiles record every original managed bank release; formal performance and Motivation services additionally retain a service-level release acknowledgment. The capability preflight rejects unknown router adapters, fused shared/routed ownership, quantized expert layouts, multi-NPU execution, and shared-expert overlap before capture. These are retained system boundaries: the evaluation establishes a single-NPU BF16 data plane, not silent fallback or support for unqualified tuples.

7 Discussion and Limitations

7.1 Stable storage is the reusable interface

Dynamic identity and graph replay conflict only when identity changes alter graph-visible allocation or execution structure. A fixed storage interface moves dynamism into contents and metadata updated at a controlled boundary. This principle may apply to other bounded working sets with a legal

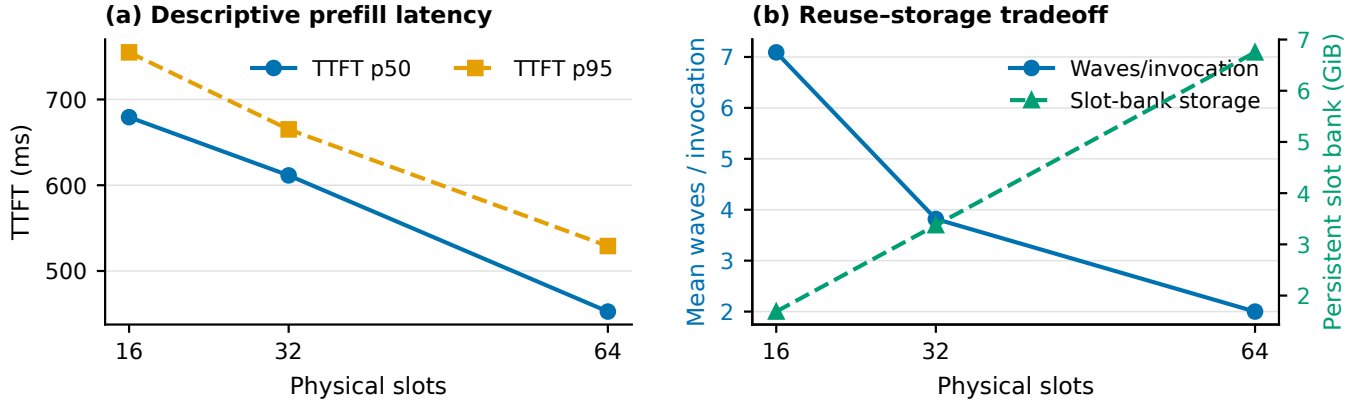


Figure 5. Online slot-capacity sensitivity for Qwen3-30B-A3B with one fresh start per point. All other serving controls are fixed, including a 14-GiB offload target, 512-MiB KV cache, 32 requests, and concurrency one on one Ascend 910B2 with BF16 and tensor parallelism one; prefix caching is disabled. More slots trade persistent HBM for fewer waves; latency values are descriptive.

staging boundary and enforceable publication/reuse order; our evidence covers only the implemented MoE path.

Address stability does not prevent stale publication or eviction; owner/version checks and stream dependencies enforce temporal correctness. Static residency sacrifices bounded offload, pointer rebinding requires graph recapture, and eager fallback pays dispatch overhead. LatchMoE fixes slots and stages replacement; its seam and fused operator are hardware-specific.

7.2 Control policy is outside the data plane

LatchMoE accepts an ordered placement plan but constrains it to the actual routed expert set. This boundary lets future caching or prediction policies reuse the fixed slots, transfer engine, wave executor, and graph checks. The current evaluation does not compare placement policies and should not be read as evidence that the default ordering is optimal. Such a study requires a shared trace, identical storage capacity, and policy-specific hit, transfer, and latency measurements on the same data plane.

7.3 Scope of the current evidence

The semantic and graph qualification covers three unquantized BF16 models on one Ascend 910B2 with tensor parallelism one. It validates native router and layer-boundary parity, exact end-to-end tokens, replay-visible addresses, overflow composition, lease/release records, and zero eager fallback for the three reported capability tuples. Performance experiments are narrower: Qwen3-30B-A3B, a 14-GiB host-offload target, serialized requests, `max_num_seqs=1`, and disabled prefix caching. The matched result shows lower TTFT for multi-wave than full-layer staging under that contract; the one-factor campaign isolates an overlap contribution. The full-resident anchor also exposes the absolute cost of

host-backed replacement. Native prefetch and legacy layered offload complete graph-mode requests but fail the exact-token comparability gate, so we retain them without using their latency. Consequently, the reported performance evidence is an internal mechanism comparison against graph-compatible full-layer staging, not a superiority claim over published offloading systems. The full-resident cost anchor is therefore central to the deployment story: the measured 17-percentage-point reduction in sampled peak HBM is purchased with a 261.28% TTFT-p50 and 309.51% TPOT-p50 penalty in this scope. We state this as a latency cost for single-card deployability, not as a universal speedup. The slot sweep has one start per point and is descriptive. None of these results establishes a general decode-speedup or multi-request serving claim.

Higher concurrency permits overlapping request leases, prefix-cache reuse adds hits across the staging seam, multi-device execution adds distributed ownership, and quantization changes layouts and operator support. None is covered by the qualified single-request host-to-device protocol.

8 Related Work

Sparse MoE inference. Sparsely gated MoE models increase parameter capacity while activating a subset of experts for each token [3, 5, 15, 16]. Systems for large MoE models also combine routing with model and expert parallelism [11, 15]. LatchMoE does not change the model, router, or expert computation. It addresses the runtime storage interface used when expert weights cannot remain fully resident on one accelerator.

Expert offloading and scheduling. Expert-offloading systems reduce host-to-device movement or its exposed cost through caching, prediction, prefetching, mixed precision, fine-grained placement, and pipelined execution [1, 4, 7, 9,

10, 19, 20, 25]. Recent CPU–GPU hybrid serving also streamloads MoE prefill weights and overlaps CPU and GPU work, while FlexGen coordinates heterogeneous placement for dense models [18, 23]. These systems decide which work or weights to place, retain, and move. LatchMoE addresses an orthogonal execution constraint: changing a cache decision must not change the weight storage bound to replayed expert computation. Its placement interface therefore feeds fixed slots and versioned leases instead of becoming part of the captured ABI.

Graph-based inference execution. Nimble compiles dynamic neural-network execution [17], while FlashInfer provides graph-oriented mechanisms for customizable LLM inference [26]. GraCE uses compiler transformations to increase graph coverage [6]. vTensor decouples computation from physical-page management while exposing a continuous virtual address [24], while ECHO makes dynamic KV-cache eviction and recall graph-friendly [12]. LatchMoE instead manages executable weights selected by a captured router and ordered against transfer and MLP computation. An eager seam exposes a fixed slot/map ABI to the downstream graph.

Positioning. LatchMoE joins dynamic placement with graph replay by separating logical ownership from graph-visible allocation and checking each ownership interval as a lease. UVA alone supplies neither bounded HBM replacement nor ready-before-publication and release-before-reuse ordering.

9 Conclusion

LatchMoE makes routing-driven expert replacement compatible with graph replay by virtualizing dynamic logical owners over address-stable physical slots and ordering each ownership interval with a versioned lease. Its captured-router, eager-staging, captured-MLP seam and capacity-bounded waves pass semantic, graph, and lifecycle qualification across three materially different MoE models. On the scoped single-NPU, concurrency-one Qwen3 workload, bounded waves change paired TTFT by -35.13% in the matched full-layer comparison, and isolated overlap changes it by -27.87%; broader concurrency, quantization, and multi-NPU claims remain outside the qualified boundary. These reductions are a capacity–latency tradeoff rather than a universal speedup: against full residency in Q2, the same workload lowers median sampled HBM peak from 97% to 80% while increasing TTFT and TPOT by 261.28% and 309.51%, respectively.

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